Passage #11: February 1995 Questions 1-6

It is a fundamental tenet of geophysics that the Earth's magnetic field can exist in either of two polarity states: a "normal" state, in which north-seeking compass needles point to the geographic north, and a "reverse" state, in which they point to the geographic south. Geological evidence shows that periodically the field's polarity reverses, and that these reversals have been taking place at an increasing rate. Evidence also indicates that the field does not reverse instantaneously from one polarity state to another; rather, the process involves a transition period that typically spans a few thousand years.

Though this much is known, the underlying causes of the reversal phenomenon are not well understood. It is generally accepted that the magnetic field itself is generated by the motion of free electrons in the outer core, a slowly churning mass of molten metal sandwiched between the Earth's mantle (the region of the Earth's interior lying below the crust) and its solid inner core. In some way that is not completely understood, gravity and the Earth's rotation, acting on temperature and density differences within the outer core fluid, provide the driving forces behind the generation of the field. The reversal phenomenon may be triggered when something disturbs the heat circulation pattern of the outer core fluid, and with it the magnetic field.

Several explanations for this phenomenon have been proposed. One proposal, the "heat-transfer hypothesis," is that the triggering process is intimately related to the way the outer core vents its heat into the mantle. For example, such heat transfer could create hotter (rising) or cooler (descending) blobs of material from the inner and outer boundaries of the fluid core, thereby perturbing the main heat-circulation pattern. A more controversial alternative proposal is the "asteroid-impact hypothesis." In this scenario an extended period of cold and darkness results from the impact of an asteroid large enough to send a great cloud of dust into the atmosphere. Following this climatic change, ocean temperatures drop and the polar ice caps grow, redistributing the Earth's seawater. This redistribution increases the rotational acceleration of the mantle, causing friction and turbulence near the outer core-mantle boundary and initiating a reversal of the magnetic field.

How well do these hypotheses account for such observations as the long-term increase in the frequency of reversal? In support of the asteroid-impact model, it has been argued that the gradual cooling of the average ocean temperature would enable progressively smaller asteroid impacts (which are known to occur more frequently than larger impacts) to cool the Earth's climate sufficiently to induce ice-cap growth and reversals. But theories that depend on extraterrestrial intervention seem less convincing than theories like the first, which account for the phenomenon solely by means of the thermodynamic state of the outer core and its effect on the mantle.

1. Which one of the following statements regarding the Earth's outer core is best supported by information presented in the passage?

   (A) Heat circulation in the outer core controls the growth and diminution of the polar ice caps.
   (B) Impact of asteroids on the Earth's surface alters the way in which the outer core vents its heat into the mantle.
   (C) Motion of electrons within the metallic fluid in the outer core produces the Earth's magnetic field.
   (D) Friction and turbulence near the boundary between the outer core and the mantle are typically caused by asteroid impacts.
   (E) Cessation of heat circulation within the outer core brings on multiple reversals in the Earth's magnetic field.

2. The author's objection to the second hypothesis discussed in the passage is most applicable to which one of the following explanations concerning the extinction of the dinosaurs?

   (A) The extinction of the dinosaurs was the result of gradual changes in the composition of the Earth's atmosphere that occurred over millions of years.
   (B) The dinosaurs became extinct when their food supply was disrupted following the emergence of mammals.
   (C) The dinosaurs succumbed to the new, colder environment brought about by a buildup of volcanic ash in the atmosphere.
   (D) After massively overpopulating the planet, dinosaurs disappeared due to widespread starvation and the rapid spread of disease.
   (E) After radical climatic changes resulted from the impact of a comet, dinosaurs disappeared from the Earth.

3. The author mentions the creation of blobs of different temperatures in the Earth's outer core (lines 34-38) primarily in order to

   (A) present a way in which the venting of heat from the outer core might disturb the heat-circulation pattern within the outer core
   (B) provide proof for the proposal that ventilation of heat from the outer core into the mantle triggers polarity reversal
   (C) give an example of the way in which heat circulates between the Earth's outer core and the Earth's exterior
   (D) describe how the outer core maintains its temperature by venting its excess heat into the Earth's mantle
   (E) argue in favor of the theory that heat circulation in the Earth's interior produces the magnetic field

4. Which one of the following statements regarding the polarity of the Earth's magnetic field is best supported by information in the passage?

   (A) Most, but not all, geophysicists agree that the Earth's magnetic field may exist in two distinct polarity states.
   (B) Changes in the polarity of the Earth's magnetic field have occurred more often in the recent past than in the distant past.
   (C) Heat transfer would cause reversals of the polarity of the Earth's magnetic field to occur more quickly than would asteroid impact.
   (D) Geophysicists' understanding of the reversal of the Earth's magnetic field has increased significantly since the introduction of the heat-transfer hypothesis.
   (E) Friction near the boundary of the inner and outer cores brings on reversal of the polarity of the geomagnetic field.

5. Which one of the following can be inferred regarding the two proposals discussed in the passage?

   (A) Since their introduction they have sharply divided the scientific community.
   (B) Both were formulated in order to explain changes in the frequency of polarity reversal.
   (C) Although no firm conclusions regarding them have yet been reached, both have been extensively investigated.
   (D) They are not the only proposals scientists have put forward to explain the phenomenon of polarity reversal.
   (E) Both were introduced some time ago and have since fallen into disfavor among geophysicists.

6. The author mentions each of the following as possible contributing causes to reversals of the Earth's magnetic field EXCEPT

   (A) changes in the way heat circulates within the outer core fluid
   (B) extended periods of colder temperatures on the Earth's surface
   (C) the creation of circulating blobs of outer core material of different temperatures
   (D) changes in circulation patterns in the Earth's oceans
   (E) clouding of the Earth's atmosphere by a large amount of dust

Passage #12: June 2006 Questions 21-27

Every culture that has adopted the cultivation of maize—also known as corn—has been radically changed by it. This crop reshaped the cultures of the Native Americans who first cultivated it, leading to such developments as the adoption of agrarian and in some cases urban lifestyles, and much of the explosion of European populations after the fifteenth century was driven by the introduction of maize together with another crop from the Americas, potatoes. The primary reason for this plant's profound influence is its sheer productivity. With maize, ancient agriculturalists could produce far more food per acre than with any other crop, and early Central Americans recognized and valued this characteristic of the plant. But why are maize and a few similar crops so much more bountiful than others? Modern biochemistry has revealed the physical mechanism underlying maize's impressive productivity.

To obtain the hydrogen they use in the production of carbohydrates through photosynthesis, all plants split water into its constituent elements, hydrogen and oxygen. They use the resultant hydrogen to form one of the molecules they need for energy, but the oxygen is released into the atmosphere. During photosynthesis, carbon dioxide that the plant takes in from the atmosphere is used to build sugars within the plant. An enzyme, rubisco, assists in the sugar-forming chemical reaction. Because of its importance in photosynthesis, rubisco is arguably the most significant enzyme in the world. Unfortunately, though, when the concentration of oxygen relative to carbon dioxide in a leaf rises to a certain level, as can happen in the presence of many common atmospheric conditions, oxygen begins to bind competitively to the enzyme, thus interfering with the photosynthetic reaction.

Some plants, however, have evolved a photosynthetic mechanism that prevents oxygen from impairing photosynthesis. These plants separate the places where they split water atoms into hydrogen and oxygen from the places where they build sugars from carbon dioxide. Water molecules are split, as in all plants, in specialized chlorophyll-containing structures in the green leaf cells, but the rubisco is sequestered within airtight tissues in the center of the leaf. The key to the process is that in these plants, oxygen and all other atmospheric gases are excluded from the cells containing rubisco. These cells, called the bundle sheath cells, surround the vascular structures of the leaf—structures that function analogously to human blood vessels. Carbon dioxide, which cannot enter these cells as a gas, first undergoes a series of reactions to form an intermediary, nongas molecule named C-4 for the four carbon atoms it contains. This molecule enters the bundle sheath cells and there undergoes reactions that release the carbon dioxide that will fuel the production of carbohydrates (e.g., sugars). Taking its name from the intermediary molecule, the entire process is called C-4 photosynthesis. Such C-4 plants as sugar cane, rice, and maize are among the world's most productive crops.

21. Which one of the following most accurately states the main point of the passage?

(A) The greater productivity of maize, as compared with many other crops, is due to its C-4 photosynthetic process, in which the reactions that build sugars are protected from the effects of excess oxygen.

(B) Because of their ability to produce greater quantities and higher qualities of nutrients, those plants, including maize, that use a C-4 photosynthetic process have helped to shape the development of many human cultures.

(C) C-4 photosynthesis, which occurs in maize, involves a complex sequence of chemical reactions that makes more efficient use of available atmospheric hydrogen than do photosynthetic reactions in non-C-4 plants.

(D) The presence of the enzyme rubisco is a key factor in the ability of C-4 plants, including maize, to circumvent the negative effects of gases such as oxygen on the production of sugars in photosynthesis.

(E) Some of the world's most productive crop plants, including maize, have evolved complex, effective mechanisms to prevent atmospheric gases that could bind competitively to rubisco from entering the plants' leaves.

22. Which one of the following most accurately describes the organization of the material presented in the second and third paragraphs of the passage?

(A) The author suggests that the widespread cultivation of a particular crop is due to its high yield, explains its high yield by describing the action of a particular enzyme in that crop, and then outlines the reasons for the evolution of that enzyme.

(B) The author explains some aspects of a biochemical process, describes a naturally occurring hindrance to that process, and then describes an evolutionary solution to that hindrance in order to explain the productivity of a particular crop.

(C) The author describes a problem inherent in certain biochemical processes, scientifically explains two ways in which organisms solve that problem, and then explains the evolutionary basis for one of those solutions.

(D) The author describes a widespread cultural phenomenon involving certain uses of a type of plant, explains the biochemical basis of the phenomenon, and then points out that certain other plants may be used for similar purposes.

(E) The author introduces a natural process, describes the biochemical reaction that is widely held to be the mechanism underlying the process, and then argues for an alternate evolutionary explanation of that process.

23. Assuming that all other relevant factors remained the same, which one of the following, if it developed in a species of plant that does not have C-4 photosynthesis, would most likely give that species an advantage similar to that which the author attributes to C-4 plants?

(A) Water is split into its constituent elements in specialized chlorophyll-containing structures in the bundle sheath cells.
(B) An enzyme with which oxygen cannot bind performs the role of rubisco.
(C) The vascular structures of the leaf become impermeable to both carbon dioxide gas and oxygen gas.
(D) The specialized chlorophyll-containing structures in which water is split surround the vascular structures of the leaf.
(E) An enzyme that does not readily react with carbon dioxide performs the role of rubisco in the green leaf cells.

24. The author's reference to "all other atmospheric gases" in line 46 plays which one of the following roles in the passage?

(A) It indicates why certain atmospheric conditions can cause excess oxygen to build up and thus hinder photosynthesis in non-C-4 plants as described in the previous paragraph.
(B) It supports the claim advanced earlier in the paragraph that oxygen is not the only atmospheric gas whose presence in the leaf can interfere with photosynthesis.
(C) It supports the conclusion that non-C-4 photosynthesis makes use of several atmospheric gases that C-4 photosynthesis does not use.
(D) It explains why carbon dioxide molecules undergo the transformations described later in the paragraph before participating in photosynthesis in C-4 plants.
(E) It advances a broader claim that oxygen levels remain constant in C-4 plants in spite of changes in atmospheric conditions.

25. The passage contains information sufficient to justify inferring which one of the following?

(A) In rice plants, atmospheric gases are prevented from entering the structures in which water is split into its constituent elements.
(B) In rice plants, oxygen produced from split water molecules binds to another type of molecule before being released into the atmosphere.
(C) Rice is an extremely productive crop that nourishes large segments of the world's population and is cultivated by various widely separated cultures.
(D) In rice plants, rubisco is isolated in the bundle sheath cells that surround the vascular structures of the leaves.
(E) Although rice is similar to maize in productivity and nutritive value, maize is the more widely cultivated crop.

26. The author of the passage would be most likely to agree with which one of the following statements?

(A) Maize's impressive productivity cannot be understood without an understanding of its cultural influences.
(B) Maize is an example of a plant in which oxygen is not released as a by-product of photosynthesis.
(C) Maize's high yields are due not only to its use of C-4 but also to its ability to produce large quantities of rubisco.
(D) Until maize was introduced to Europeans by Native Americans, European populations lacked the agricultural techniques required for the cultivation of C-4 plants.
(E) Maize's C-4 photosynthesis is an example of an effective evolutionary adaptation that has come to benefit humans.

27. The passage provides the most support for which one of the following statements?

(A) In many plants, rubisco is not isolated in airtight tissues in the center of the leaf.
(B) A rubisco molecule contains four carbon atoms.
(C) Rubisco is needed in photosynthesis to convert carbon dioxide to a nongas molecule.
(D) In maize, rubisco helps protect against the detrimental effects of oxygen buildup in the leaves.
(E) Rubisco's role in the C-4 process is optimized when oxygen levels are high relative to carbon dioxide levels.

Additional Reading Comprehension—Science and Technology Passages Line Reference Notes and Answer Key

Passage #11: February 1995 Questions 1-6

This passage begins by introducing the concept of magnetic field reversal: occasionally the Earth's magnetic field reverses itself so that magnetic "north" and magnetic "south" appear to have switched geographical places. While scientists have a general understanding of how the magnetic field itself is formed (due to the motion of molten metal in the outer core), there are a number of competing hypotheses designed to explain the actual reversal. Two of these—heat-transfer and asteroid-impact—are discussed, with the author concluding the passage by suggesting that the heat-transfer theory seems to be more probable.

1. C	(CR, Must)	16-21	"It is.....inner core."
2. E	(CR, Must)	38-49	"A more.....magnetic field."
		59-63	"But theories.....the mantle."
3. A	(SR, Must, P)	31-38	"One proposal.....heat-circulation pattern."
4. B	(CR, Must)	6-9	"Geological evidence.....increasing rate."
5. D	(CR, Must)	30-31	"Several explanations.....been proposed." *Note: The word "several" is typically defined as "more than two but fewer than many."
6. D	(CR, Must X)	30-49	"Several explanations.....magnetic field."

Passage #12: June 2006 Questions 21-27

This passage begins with a fairly broad description of maize and its impressive productivity, and then goes on to outline in great detail the process by which maize is able to achieve this high level of productivity. Through a special type of photosynthesis, maize (and C-4 plants) is able to keep excess oxygen from inhibiting sugar production by housing its sugar-producing enzyme in airtight structures and thus achieving a higher rate of growth. Test takers had the most trouble with question #24.

21. A	(GR, Must, MP)	All	
22. B	(SR, Must, O)	Paragraphs 2 and 3	
23. B	(CR, Must)	30-38	"Unfortunately, though……impairing photosynthesis."
24. D	(SR, Must, P)	45-54	"The key……it contains."
25. D	(GR, Must)	57-61	"Taking its……productive crops."
26. E	(GR, Must, AP)	1-18	"Every culture……impressive productivity."
27. A	(GR, Must)	36-38	"Some plants,……impairing photosynthesis."